
GEO 455/555: Webmapping

Department of Geography & Environmental Science, University of Wisconsin-La Crosse
Spring 2025, 3 Credits

Instructor:	Gargi Chaudhuri, PhD	Class Location:	Cartwright Hall 326
Email:	gchaudhuri@uwlax.edu	Lecture Day/Time:	Tues/Thurs-9:55-10:50AM
Office Hours:	T/W/Th- 8:30-9:30AM	Lab Location:	Cartwright Hall 331
Office Location:	Cartwright Hall 330-C	Lab Day/Time:	Tues/Thurs-11:00-11:55AM

1 COURSE DESCRIPTION

Online (web based) maps are increasingly becoming the preferred medium over traditional paper maps. Rather than being static, web maps are interactive and allow users to load information that they want, zoom in, zoom out, focus on an area of interest and even change the map presentation format. Creating and designing effective online maps is becoming both interesting and challenging. The aim of this course is to introduce students to the use and development of web based mapping tools to support information sharing, geodesign and decision making. The course emphasizes practical applications of Web GIS through Web scripting and user interface design using a variety of applications and application programming interfaces (API). The important elements of this course include:

- Scripting of Web based interfaces (HTML, CSS, JavaScript)
- User interface (UI) design (layout, navigation, browser/device limitations)
- Application Programming Interfaces (API) for web mapping such as Leaflet for JS

1.1 Prerequisites

GEO 418 or GEO 405. It will be assumed that students will know the following: Desktop computers and Windows operating system; Web Browsing, Microsoft (MS) Excel, MS Word, MS Powerpoint, save a file to a given drive, create, name and rename files and folders, drag and drop files, send an email attachment, know/how of Canvas, know how/when to log-off/ shutdown, find a file or folder, know when it is appropriate to use 'Save as' instead of 'Save', download files from the web to a given folder, move files between drives, Zip/Unzip a file, use the start menu to locate and start a software, copy, paste and delete files, Select & print to a printer from Windows application, make a new folder in an existing folder, place an image in Microsoft Word/PowerPoint, and use "Help" from within a Windows application.

2 READINGS:

- Text Book: Dorman, Michael (2022), Introduction to Web Mapping, CRC Press.
- Additional reading: Muehlenhaus, Ian (2014), Web Cartography, CRC Press. Referred as IM in the schedule.

3 STUDENT LEARNING OUTCOMES

A successful GEO 455/555 student will learn:

- Prepare data for use in Web interface.
- Identify suitable Web Mapping methods to use for various project types.

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- Design, develop and deploy interactive mapping tools.
 - Identify interface usability.

3.1 Expectations from students

To be successful in this course, a student must

- Read assigned readings and go through the lecture materials regularly
- Listen to lectures carefully and take notes over written materials.
- Attend class regularly.
- Submit completed assignments, exams, homeworks, final projects and any other class activity within due dates.
- Ask questions and discuss topics within the lecture session. Clarify issues within class itself while the topic is being taught.
- **Graduate Students** are expected to produce better quality of the work for full grades.

4 ASSIGNMENTS

- **Lab Assignments:** You need to complete all lab assignments and submit in required format by its due date to be considered for full grades in each assignment. If you fail the lab assignments, you will fail the course. Late submission of any assignment is strongly discouraged. However, turning a lab late is encouraged over not doing it at all. Late submissions will be accepted until the last day of classes. **One week late submission will lose 25%** of their value and **more than one week late submission will lose 50%** of their value if more than one week late. In case of sickness or emergency, appropriate proof should be provided during late submission otherwise points would be deducted. You will have enough time to complete each assignment within assigned lab time with help from me and/or lab assistant. If you choose not to complete your work within lab time, it's your responsibility to finish on time by yourself.
- **Exercises:** There will be a number of exercises that you will be provided based on some lecture materials.
- **Review presentations:** Students will be assigned review materials to critically evaluate and present in class.
- **Exams:** There will be **two exams on the dates listed in the syllabus** and are designed to assess your comprehension of the presented materials. Two exams will be based on lecture materials and readings and will be named as Mid-term and Final exams. Make-up exams will be given **ONLY** during exceptional circumstances, such as illness or university-approved event, and proof will be required. In cases where you have a scheduling conflict with a university-approved event, it is **YOUR** responsibility to notify the instructor at least two weeks before the exam. In the case of emergencies, it is **YOUR** responsibility to notify the instructor that you will be unable to attend, preferably before the exam or within 24 hours after the exam.
- **Final Project:** More information about the final project can be found in Final Project module in canvas. In the final project, students are required to complete a final web mapping project that will demonstrate their skill sets and tools learned in this class. For selecting the thematic focus/topic for the web mapping project student will discuss with the instructor. During the various lectures and lab exercises students will be shown multiple online/secondary data sources to procure spatial datasets that could be used for this project. The whole process of final projects takes last few weeks, and students are guided by few milestones which help them accomplish their task on time. **The final project submission items are not eligible for late submission.** If you fail to turn in any of the items by the due date then you will not receive any points for that item.

- **For graduate students:** In addition to the items above, graduate students are required to write an academic paper based on their final project. Graduate students are strongly suggested to focus their final project topic within the context of their thesis work.

5 ATTENDANCE & AUTHORIZED ABSENCES

Students earn 1 point for attending class. Unexcused absences earn -1 point. Authorized absences documented in a timely manner under university policy, including university-sponsored events, are not penalized and therefore are excluded from the attendance calculation rather than counted as present or absent. Students with authorized absences remain responsible for missed material and any make-up work assigned. Please email the instructor if you plan/need to miss a class (lecture and/or lab) due to a specific reason. If you do not inform and miss a class (lecture and/or lab) half of the point will be deducted for each section.

6 FORMAT & COURSE ACCESS

This is a face-to-face course. You need to be in class to attend lectures and work on lab assignments. You can ask questions and clarify doubts with the instructor in class during your session. You will need your UWL NetID to login to the course from the CANVAS homepage. Course materials will be available online, including lectures, lab assignments handouts, video demonstrations for assignments, quizzes, assignment dropboxes, and other miscellaneous instructions. The lab assignment data will be available via Geography server folders. Access to course materials in CANVAS will cease after the term ends. If you wish to archive materials for your personal records or portfolio you should do so as you progress through the course. As a general rule, you should always save local copies of course-related work. To avoid disasters, you should also save important files to external media or cloud storage.

6.1 Technical Support

For tips and information about Canvas visit the Canvas Guide for Student. You can also contact the ITS Support Center at (608) 785-8774 or helpdesk@uwlax.edu for questions about Canvas or any other technological difficulties. The hours for ITS are Monday through Thursday from 7:30 am to 6:30 pm, and Friday from 7:30 am to 4:30 pm, Central Time. For GIS software and computer issues in the lab please contact the GIS Lab Administrator Steve Fulton.

7 GRADING

7.1 SCHEME (in %)

A 93 – 100 | AB 88 – 92 | B 83 – 87 | BC 78 – 82 | C 70 - 77 | D 60 - 69 | F 59 & below

7.2 WEIGHT

Lab Assignments -	Assignments	40%
Exams -	Module Quizzes -	20%
Final Project -	Proposal, draft presentation, final presentation -	25%
	Graduate Students* - includes draft paper and final paper	
Class Participation -	Exercises, Attendance, Reviews -	15%

8 STUDENT SUPPORT SERVICES

8.1 How to contact the instructor

If you have any issues talk to your instructor during class time or office hours. Check for email and office hours in syllabus. Do not completely rely on emails. If you have to send an email, email with clear subject, course and section number, use proper introduction, body of the email should explain the issue properly, and sign off with your name.

8.2 Expectations for graded work

Generally, I return graded assignments with individualized feedback, if needed, within 21 days from the due date of the assignment. I will notify you if I am unable to grade the work within the 21-day timeframe, and will identify a revised return date. If you submit work after the due date, it may not be returned within 21 days. The grades for any work that is graded electronically, such as scanned examinations or automated quiz, will be accessible to you within 21 days of the due date. If you submit electronically graded work after the due date, it may not be accessible within 21 days. Your graded coursework will be returned in compliance with FERPA regulations, such as in class, during my office hours, or via the course management system through which only you will have access to your grades.

8.3 UWL Syllabus Policy Information & Statements

UWL encourages students to know the campus' important policies and statements which can be found on the Syllabus Information website.

- Academic integrity and misconduct
- Classes during inclement weather
- Religious accommodations
- Sexual misconduct
- Student course and faculty-related concerns, complaints, and grievances
- Student survey on instruction (LENS)
- Students with accommodation needs
- University class attendance policy
- Veterans, active military, and military-connected

8.4 Academic Services and resources at UWL

Please check here to find out more about Academic Services and resources at UWL

8.5 PRO@UWL

PRO@UWL (Progress Report Online via Navigate)/Student Success Policy - If I notice that you are experiencing difficulties early in the semester (e.g., low assignment scores or limited participation), I may provide you feedback through Navigate, UWL's success system, and you will receive notification indicating that I have entered feedback. I encourage you to meet with me and/or utilize helpful campus resources listed on UWL's Student Success website.

8.6 Inclusive Excellence

UWL's core values include "Diversity, equity, and the inclusion and engagement of all people in a safe campus climate that embraces and respects the innumerable different perspectives found within an increasingly integrated and culturally diverse global community." If you are not experiencing my class in this manner, please come talk to me about your experiences so I can try to adjust the course if possible.

8.7 Name/Pronouns

I will do my best to address you by a preferred name or gender pronoun that you have identified. Please advise me of this preference early in the semester so that I may make appropriate changes to my records. UWL has a preferred name policy and UWL's Pride Center is available for additional assistance.

8.8 Academic Success and Overall Health

At UWL, we support your academic success and overall health. We know that students often experience a range of stressors that can impact learning and well-being. If you or someone you know is experiencing mental health concerns, or could benefit from effective academic strategies, there are free and confidential resources available to enrolled students through the Counseling & Testing Center (CTC). To learn more, visit CTC's website or call 608-785-8073.

8.9 Statement for Courses Limiting Use of AI

Artificial intelligence (AI) tools and resources used for creating text, images, computer code, audio, or other media, may be used in certain circumstances in this class. The specifics of when, where, and how these tools are permitted will be included with each assignment. Guidance for attribution/citation will be provided by your instructor. Any use of these tools other than where/how indicated is a violation of course expectations and is subject to UW-La Crosse's academic misconduct policy. Students are responsible for verifying the accuracy and appropriateness of content composed by AI. If you are in doubt as to whether you are using a tool inappropriately in this course, you are encouraged to discuss your situation with your instructor.

8.10 Regular and substantive interaction (RSI)

In this course, regular and substantive interaction between the instructor and students plays a crucial role in student achievement. I will be incorporating various forms of interaction that not only meet the requirements set by the US Department of Education and the UWL Faculty Senate policy, but also aid in student achievement. Your active participation and contributions are also essential to your learning. In partnership with your efforts, I will :

- be available during scheduled student hours as stated in the syllabus
- share information about the course materials including key information, explanations, examples, and resources via in-person, and/or text-based lectures
- engage in discussions about course content
- provide group or individual feedback on assignments
- respond to questions about the course sent via email within 24 hours
- post announcements pertaining to the course content and activities on Canvas, as needed
- monitor your academic progress and communicate concerns, as needed

9 SCHEDULE

Details about due dates and any changes/updates in the schedule will be announced in class and/or Canvas.

Week 1: 1/27 and 1/29

1/27

Lec#0: Course Mechanics
Lab#0 set-up – Github account and code editor
Homework: Watch the EdX CS50 Lecture 8
Sign-up for webmap review

1/29

Lec#1: Basics of Webmapping technology
Reading: IM-Ch.12

Week 2: 2/3 and 2/5

2/3

Lec#1: HTML & CSS
Lab#1 ; Ex#1 & #2
Reading: Ch. 1 & Ch.2
Webmap review

2/5

Continue with lecture,
Work on lab#1, Ex#1 & #2
Webmap review

Week 3: 2/10 and 2/12

2/10

Lec#2: JavaScript
Lab#2; Ex#3
Reading: Ch.3
Webmap review

2/12

Lec#3: Webmap Elements
Work on Lab#2 and Ex#3
Reading: IM Ch.3
Webmap review

Week 4: 2/17 and 2/19

2/17

Lec#4: JavaScript Interactivity
Lab#3; Ex#4
Reading: Ch.4
Webmap review

2/19

Continue lecture
Work on Lab#3 and Ex#4
Webmap review

Week 5: 2/24 and 2/26

2/24

Lec#5: Webmap composition and layout
Discuss Mid-term Exam
Reading: Ch.5
Webmap review

2/26

Watch Alex Tait video on Map Composition
Read Anatomy of Webmap & take quiz
Webmap review
Read IM-Ch.4

Week 6: 3/3 and 3/5

3/3

Mid-term Exam
Lab#4

3/5

Lec#6: Leaflet
Work on Lab#4
Reading: Ch.6
Webmap review

Week 7: 3/10 and 3/12

3/10

Lec#7: GeoJSON

Lab#5; Ex#5

Webmap review

3/12

Work on Lab#5 and Ex#5

Webmap review

Week 8: 3/17 and 3/19: Spring Break – No Classes

Week 9: 3/24 and 3/26

3/24

Lec#8: Thematic Visualization

Lab#6

Reading: IM-Ch.9

Webmap review

3/26

Discuss final project

Work on Lab#6

Webmap review

Week 10: 3/31 and 4/2

3/31

Work on Lab#6

Webmap review

4/2

Lec#9: jQuery

Lab#7; Ex#6

Reading: TBD

Webmap review

Week 11: 4/7 and 4/9

4/7

Lec#10: API and Real-time data integration

Work on Lab#7

4/9

Lab#8

Final project proposal due

Week 12: 4/14 and 4/16

4/14

Lec#11: Animation

Reading: IM-Ch.10

Work on Lab#8

4/16

Work on final project

Week 13: 4/21 and 4/23- Work on final project

Week 14: 4/28 and 4/30

4/28

One-on-one draft presentation / demonstration

4/30

Work on final project

Week 15: 5/5 and 5/7

4/30

Final Project Presentation – in class

5/2

Final Project Presentation – in class

Final Exam: May 14 (Wednesday) - 10:00 AM - Noon